

Beyond Riding: Passenger Engagement with Driver Labor through Gamified Interactions

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Modern cities increasingly rely on ridesharing services for on-demand transportation, which offer consumers convenience and mobility across the globe. However, these marketed consumer affordances give rise to burdens and vulnerabilities that drivers shoulder alone, without adequate infrastructures for labor regulations or consumer-led advocacy. To effectively and sustainably advance protections and oversight for drivers, consumers must first be aware of the labor, logistics and costs involved with ridehail driving. To motivate consumers to practice more socially responsible consumption behaviors and foster solidarity with drivers, we explore the potential for gamified in-ride interactions to facilitate engagement with real (and lived) driver experiences. Through nine workshops with 19 drivers and 15 passengers, we surface how gamified in-ride interactions revealed passenger knowledge gaps around latent ridehail conditions, prompt reflection and shifts in perception of their relative power and consumption behaviors, and highlight drivers' preferences for creating more immersive and contextualized service experiences, and identify opportunities to design safe and appropriate passenger-driver interactions that motivate solidarity with drivers. In sum, we advance conceptual understandings of in-ride social and managerial relations, demonstrate potential for future worker advocacy in algorithmically-managed labor, and offer design guidelines for more human-centered workplace technologies.

Additional Key Words and Phrases: Labor, Algorithmic management, Gamification, Rideshare Platforms

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1 INTRODUCTION

Platform-based gig labor occupies a growing proportion of the global workforce. Ridehail services in particular (e.g., Uber, Lyft), are crucial to advancing urban mobility [170], work access [132], and in stimulating local economies [55, 86]. But alongside increasing consumer adoption, scholars raise concerns around the physical [29, 101, 136, 159], financial [20, 150, 169] and psychosocial [8, 13, 134, 148] working conditions. Legal and critical scholars, for instance, challenge platforms' abuse of power through algorithmic management and control [76, 95], which are left unchecked by labor regulations and their consequences remain unaddressed due to inadequate social safety nets [39, 58].

Besides platforms, consumers also shape worker conditions in significant ways: as *managers* of individual interactions when they evaluate and expect service quality of workers [123, 171] and as *labor market regulators* who hold significant collective political power [59, 60]. Platforms often intentionally campaign to amass such consumer support to bypass regulatory constraints imposed by legislatures, such as with Prop 22 [27, 149]. But despite their influence, consumers remain largely unaware of laborers' harsh working realities – Pew Research found that nearly half of Americans to have never heard of ongoing debates around the classification of ridehail drivers [2].

In addition to cultivating political alignment with consumers [52, 149], platforms also intentionally design interfaces that suppress information-sharing among users [128], which constrain the capacities of workers and consumers to engage in mutual support [166]. In attempts to alleviate the resulting information asymmetries, a burgeoning body of HCI studies engage with workers such as ridehail drivers to expose the hidden and undocumented risks of gig labor (e.g., pay rates [19], deactivation [122], tax or tort liability [28], consumer misbehaviors [137]), offering an array of worker-centered tools to collectivize and resist [19, 20, 70, 75, 168]. Workers also leverage a host of commercial

tools to self-track data, as a means to manage their own accountabilities [62]. These systems sought to empower laborers through interactions that directly interface with workers, but workers occupy relatively vulnerable positions vis-a-vis platforms, making them powerless against harmful platform tactics and insufficient regulatory infrastructures. Consumers, on the other hand, hold significantly more capacity, resources and power to and influence worker rights and conditions [59, 107]. But it remains unknown how technologies can surface driving labor in ways that *motivate*, mobilize and catalyze consumers to advocate for and practice more socially-conscious consumption behaviors.

In many on-demand gig work settings, the potential to build solidarity between consumers and workers is limited by the lack of opportunities for social interaction between stakeholders, due to physical (e.g., crowdwork) or temporal distance (e.g., food delivery, petsitting). The casual, co-located setting of ridehail creates a unique space where passenger-driver pairs can engage in conversations that foster mutual understanding, rapport and support. However, the temporal constraints and social boundaries of rides – i.e. their fleeting duration and awkwardness of engaging with strangers – can still prevent both parties from initiating more meaningful and transformative exchanges. Raising the question: what forms of technological probes can motivate passengers to overcome these barriers and engage in more substantial and in-depth in-ride social interactions with drivers?

Gamification represents one approach for motivating an audience to engage with serious but sensitive prosocial causes – e.g., gender-based violence [129], interpersonal racism [151] and HIV prevention [64] – particularly within time-constrained contexts such as platform-mediated rides. While platforms leverage gamification for more managerial purposes [63], transformational and persuasive games (achieved through mechanisms of “*embedded*” messaging, interactive narratives, etc.) offer players immersive spaces to learn about or experience driving conditions over a short time span without being subjected to personally vulnerable positions. This study explores the potentials of game-based interventions as boundary objects for initiating, mediating and potentially transforming consumer discourse around the obscured realities of ridehail driving conditions. Following relevant game design techniques and frameworks for transforming player attitudes on serious social issues [32], we worked with ridehail drivers and passengers over a series of co-design sessions to explore whether gameplay interventions can mobilize passengers to engage with, understand, care and advocate for latent labor issues of ridehail labor.

RQ 1 Which forms of gamification can effectively embed latent ridehail concepts and experiences that drivers prioritize sharing with passengers?

RQ 2 How can playable interventions motivate scalable passenger engagement with, understanding of and advocacy for the working conditions of ridehail drivers?

We took an iterative design process to approach these inquiries, beginning with goal delineation that combined broader insights from related ridehail studies and formative interviews (§5.1), followed by implementation of six gamified prototypes (§4) and a series of nine co-design workshops to gather driver and passenger feedback (§5.2). Our results show (1) passenger knowledge gaps around (latent) ridehail conditions (e.g., pay, logistics, rating pressures, long-term consequences) and early evidence for gamified interventions to mediate in-depth driver-passenger interactions and solidarity (2) driver preferences for more immersive and personalized in-ride passenger-facing play experiences and (3) design tradeoffs for shaping future passenger-driver interactions that motivate passenger-led advocacy for ridehail labor. We close by discussing prospective implications for design, advocacy and technology.

2 RELATED WORK

We overview documented stressors of ridehail driving (§2.1), existing approaches to advocacy (§2.1.2), potential for consumers to initiate influence individual service encounters (§2.2.1) and regulations more collectively (§2.2.2), as well as how in-ride games (§2.3.3) can transform (§2.3.2) rather than coerce and manufacture driver labor (§2.3.1).

2.1 App-based Ridehail Services

App-based ridehail services have proliferated in the US market since their introduction more than a decade and half ago. With more than 36% of the US adults having used ridehail services [79], Uber alone captured \$37.28 billion [74] in 2023 while Lyft netted \$4.4 billion. Together with delivery work, NBER found transportation services such as ridehail to comprise the largest component of the gig workforce, as measured by the number of workers [50]. Below, we summarize existing studies documenting subpar conditions of ridehail driving, as well as prior HCI and CSCW investigations that sought to counteract exploitative platform practices via interventions such as worker data-sharing.

2.1.1 Latent Stressors, Labor and Power Asymmetry in Ridehail Driving. Despite increasing consumer adoption of app-based ridehailing services, drivers' earnings [19, 122], job quality [51, 162] and work conditions [18, 29, 100, 139, 142] in general have declined – leaving them stressed [5, 10, 130], overworked [1, 77] and seeking strategies to resist platform control [23, 104]. Researchers extensively documented how algorithmic management and intense competition creates immense psychological stress for drivers [5, 130] – who also deal with material stressors such as low pay [19, 67], hidden health and safety risks from accidents on the road [29, 100, 139, 148], violence from passengers [101, 142], fatigue [77], job precarity [120, 154] as well as more latent long-term consequences, including musculoskeletal and urinary disorders [5, 18]. However, many harmful, latent and delayed effects [18] remain unobservable to passengers, and computing scholars studying labor overlooked how consumers are critically elevated to positions of relative (but uninformed) managerial power, due to their objectives to center and impact worker priorities [169]. Specifically, platforms leverage strict 5-star rating thresholds to impose reputational pressures [104] that exploitatively discipline drivers [25]. The rating mechanism is central to platform operations [127, 128] and enable (unrealistic) consumer expectations of service quality, including requirements to maintain “friendly”, “positive” and “respectful” attitudes, regardless of how passengers themselves behave [26]. Literature from organizational science [96, 104] show how dominant ridehail platforms such as Uber design interactions that create greater power imbalance between consumers and workers to effectively “launder” latent emotional [22], psychological [113] and immaterial [123] labor from ridehail drivers [155]. Such platform designs empower passengers without providing sufficient context on latent driver labor, which can cause uninformed and/or unconscious consumption behaviors.

2.1.2 Technological Advocacy for Ridehail. Scholars at the intersection of HCI and labor studies made several attempts to curb the harmful impacts of algorithmic management [76, 95] and information asymmetries [128] by actively involving workers to participate in the co-design and development of alternative technological (data) probes and interventions. Stein et al. [143] imagined alternative data uses and more plural sociotechnical infrastructures with drivers. Zhang et al. [168] invited drivers to propose algorithmic imaginaries that center worker well-being. Hsieh et al. [68] worked with multiple stakeholder groups to reveal (mis)aligned objectives to advance technology, service and public infrastructure, including more accurate public perceptions of workers. Beyond co-design, recent studies showed the promise for data probes [168], data-sharing tools [19, 20, 70] and collectives [69] to elevate worker priorities as well as labor regulations and protections [19]. While these proposed interactions demonstrated shared worker motivations, they require effortful data contributions from drivers, many of whom labor for long hours to balance financial needs [162] with unstable job opportunities [120], making it impractical for them to engage in additional (uncompensated) interactions.

2.2 Role of Consumers in Ridehail Labor

Given the capacity constraints of workers, we turn to the untapped potential of mobilizing consumers as users of worker-advocacy technologies. First, we examine the role of passengers in managing frontline ridehail service (quality) before showing how as a collective, consumers can offer workers *political leverage over platform decisions*.

2.2.1 Passengers as Co-managers of Ridehail Service. Besides the harmful effects of algorithmic management, **consumers** also play a crucial role in rating systems [60], which are intentionally designed to provide safe, reliable and quality service to paying users [4]. Initially, maintaining quality service was necessary for platforms to amass a sizable consumer base and establish market position [104]; but over time, strict deactivation thresholds [6] became strategic political tools to forge political alliances with consumers, who can help defend against (e.g., Prop 22 [149]) or even preempt [30] emerging regulations that threaten platform operations. Combined with platform nudges [145] and mechanisms (e.g., tipping [40]), the resulting reputational pressures discipline drivers into performing unpaid service – e.g., maintaining a friendly mood [26] or clean car [4], playing comfortable music [123], dressing professionally [104], offering water, snacks or phone chargers [137]. The relative empowerment of passengers also enables abuse and disrespect towards drivers [92, 96, 104], not to mention legally questionable practices – e.g., sneaking in too many people, or alcohol [104]. In response, drivers must undertake yet more emotional labor to absorb or resist unjust treatment [165], or to proactively educate passengers about consequences through in-car fliers or conversation [128].

By redistributing, decentralizing and delegating **managerial oversight** to individual consumers, platforms (1) sustain competitive pools of drivers [6, 137] and (2) sidestep employer obligations when they circumvent the need to establish explicit service rules [54, 104, 123, 149]. Prior work highlight the power of algorithmic management [69, 143], but overlooked how *consumer* data contributions (e.g., ratings) constitute ultimate sources of this power [54]. Consequently, drivers are coerced into unpaid emotional labor, akin to traditional frontline services (e.g., call centers [104]) and more face-to-face, yet increasingly algorithmically-managed sectors like hospitality [140, 141].

2.2.2 Collective Consumer Influence on Policy & Regulation. Crucially, how drivers and consumers perceive a platform’s labor conditions can influence their use, recommendation of [7], or even retaliation against [165] it. Competing companies (e.g., Lyft [104]) with less coercive practices can attract socially conscious consumers [7], threatening dominant platforms. Thus, consumers hold scalable political power [61] that platforms actively seek to influence.

Thelen [149]’s comparative political analysis showed how Uber built coalitions with consumers to shield [31] against regulatory challenges. When London refused to renew their taxi service license in 2017 for its “*lack of corporate responsibility*” [17], Uber successfully mobilized half a million users within one day using its “*tried-and-tested tactic of asking customers for help when it locks horns with regulators*” [34, 42] – resulting in license reinstatement. Healy et al. [61] explored how workers might “*gain support from consumers they serve*” [60], to influence not only individual service interactions (managerially) but also broader platform accountability through collective political power [59].

However, consumers currently lack awareness of the need to further regulate ridehail platforms: while consumers with knowledge about driver classification were 20% more likely to support additional regulation, over 40% of American adults had never heard about the debate around the non-employee status of ridehail drivers in 2021 [2, 58] – shortly after Prop 22 overturned Assembly Bill 5. Meanwhile, 2022 surveys show conflicted consumer opinions about platform-based labor, especially regarding hidden aspects of working conditions – e.g., long-term consequences on career [60]. But Healy and Pekarek [60] also revealed a cooling trend in consumers’ favorable perceptions of platforms, uncovering an opportunity to awaken their citizen identities [31] and mobilize consumer solidarity for workers through their concern for causes such as social responsibility [54], environmentalism [7] or philanthropy [167].

2.3 Gamification to Motivate Citizen Engagement

Prior work has attempted to design for consumer awareness, compassion and action [94] in the gig economy, but educating and motivating the geographically-dispersed population of ridehail passengers can be challenging, especially if constrained by the short time span of a single ride. Below, we summarize how platforms design algorithmic games to incentivize driver labor, the potential for gamification to rapidly mobilize passengers for driver advocacy, as well as the state of the art on games designed for, or related to ridehail.

2.3.1 Gamification: Manufacturers of Consent vs Drivers of Transformation. Platforms employ several gamification techniques to psychologically trick and coerce labor [22, 117, 135] – which drivers resist [157]. Mechanisms include Quests and Challenges that yield monetary prizes, as well as more symbolic reward structures such as badges, points and status programs [25, 157]. But while such tactics motivate initial driver engagement, managerially-imposed gamification also take away their “sense of consent” and agency to set practical boundaries with work [84, 109]. Worse, the combination of opaque but rapid algorithmic rewards result in paradoxical senses of stressful and frustrating motivation that cannot be resolved by human managers [84, 128]. Such managerial gamification [109] leverage phenomena such as “income targeting” from behavioral economics to trick drivers into “grinding” away for long hours [134].

On the other side of managerially controlling workplace games is the possibility of more humanistic and transformative game play [32, 37], where players engage in autonomous, and growth-oriented games that guides them to grow, make meaning and learn – such as when drivers practice oppositional play to resist workplace games [22, 157]. Transformational game design [32, 152] adopts the latter, less managerial rhetoric and opens up immense opportunities to embed principles from various disciplines (e.g., asymmetric paternalism from behavioral economics [21], incentive compatibility from mechanism design) to develop games that illustrate alternative choice architectures that motivate better player decisions and growth that can extend beyond the game environment into real-life choices.

2.3.2 Games to Transform Passenger Engagement with Driver Experiences & Labor. A key barrier to approaching and discussing the challenges and mechanisms of ridehail work is the sensitive and private nature of financial and emotional vulnerabilities that drivers experience [132], which can prevent consumers from probing about hidden driving labor and logistics (§2.1). Gamified environments and gameful designs [78] present an opportunity to build safe and inclusive spaces that can rapidly engage player awareness of sensitive [83, 151], complex, and overlooked topics [121].

Games have functioned as a medium for promoting critical thinking and social consciousness around various pressing societal issues, ranging from racism (e.g., *SimCity* [47]) to colonialism (e.g., *Civilization* [110]) to capitalism (e.g., *Animal Crossing* [15], *World of Warcraft*, *Second Life* [57]), including specific dimensions such as immaterial labor (e.g., *Mario* [126]). Persuasive games intentionally leverage techniques like procedural rhetoric (the use of rules, mechanics and decisions) to model and portray social systems [14], embedded approaches (e.g., distancing and intermixing) to address controversial topics, and immersive methods like narrative role-play and role reversal [90, 164] to affectively engage players in more marginalized and constrained perspectives [33, 46]. Temporal features such as immediate feedback and timed challenges/pressure [9] are shown to foster better and quicker user engagement [11]. In ridehail, gamification offer ideal opportunities to rapidly engage players by

- (1) psychologically distancing the driver and passenger, who can then feel more
 - (a) safe and motivated to explore driving conditions through immersive and (temporally) challenging spaces
 - (b) socially comfortable to discuss legitimate and lived driving experiences with drivers
- (2) simulating gamification tactics that platforms impose to exert psychological control.

2.3.3 *State of the Art for Ridehail Games.* Presently, we are aware of only one system occupying the space of in-ride interactive games: the [Play Octopus Network](#), which markets itself as the world’s “*largest rideshare advertising network*” and provides drivers in-car tablets with advergames such as trivia that primarily function to engage passenger attention for generating advertising revenue. Popan et al. [116] attempted to leverage gamified role-playing to mobilize drivers, which presupposes additional effort and capacity that most drivers lack. A few browser-based games also exist that intend to capture a broader audience, including the *Uber Game* [45] (which won an Online Journalism Award) and *Cherry Picker* [12], but we remain unaware of any games designed for the situated context of a ride.

3 DESIGN CRITERIA FOR GAMIFIED IN-RIDE INTERACTIONS

To effectively convey hidden driving labor and conditions to passengers, we identified relevant criteria from game design and heuristics to support our study goals – described below and summarized in Table 1. Through pilot interviews and goal refinement (detailed in §5.1), we also identified three additional ridehail-specific criteria in §3.3.

3.1 Game Design Heuristics

3.1.1 *Replayable.* One metric for assessing a game’s engagement potential is the player’s desire to play again [35]. Replay can enhance learning and understanding of game content [99], which is crucial first step to our goal of engaging passenger interest in ridehail labor conditions. Replays also promote social interaction among players (e.g., through discussion of its content) [48], which further supports our objective of mobilizing consumers.

3.1.2 *(Timed) Challenge.* Another standard playability heuristic in both video and mobile games [35, 89] revolves around the presence of a meaningful challenge – i.e. the level of difficulty that players must overcome to reach a winning condition – which is central to creating enjoyable experiences. Malone [105] defined that challenging games must contain “*a goal whose outcome is uncertain*”, while Desurvire and Wiberg [36] also considered temporal factors: “*well-paced challenge(s) that makes the game worth playing*”. In ridehail games, timed challenges additionally simulate drivers’ realistic time constraints [5]. Despite incorporating challenges, we refrained from incentives such as leaderboards or challenge invites to contacts (e.g., friends or family) since they can trivialize driver vulnerabilities [129, 160].

3.2 Embedded Design Strategies

Kaufman et al. [83] recommends “*stealthily*” embedding persuasive messaging to ease players into receiving intended messages. We overview below ways to leverage embedded design strategies – i.e., intermixing, obfuscating and distancing – to effectively communicate latent ridehail driving conditions.

3.2.1 *Intermixing.* Interspersing thematic on-message (e.g., ridehail content) and playful off-message (e.g., narrative content) material guides players into subconsciously internalizing a game’s intended themes – offsetting initial player discomfort and reservation when presented with emotionally-taxing topics [114]. Passengers who resist explicitly acknowledging the effects of their consumption behaviors may prefer interwoven content over overt content delivery.

3.2.2 *Obfuscating.* To further bypass players’ psychological defense, obfuscation conceals the persuasive intent of purposive games, reorienting player attention toward apparent game mechanics and objectives. The technique has been applied to topics such as bias against women in STEM [49], complex social identities [82] and health advocacy [83]. By introducing serious and persuasive material messages covertly, obfuscation helps passengers avoid feeling pressures from their consumer identities (or associated managerial powers) while still provoking critical reflection from the user.

3.2.3 *Psychological Distancing through Immersive Fictional Narratives.* Narrative fiction is an effective medium for communicating complex and sensitive social experiences, including gender-based violence [129], interpersonal racism [151], healthcare [71], and climate change [44]. Immersive narrative framings of social experiences also facilitate reflection [72], empathetic growth [53, 106] and prosocial behaviors [103, 163], allowing us to create safe spaces for players to explore sensitive driver topics in the first-person without directly experiencing harmful or disturbing work conditions. *Interactive* narrative fictions further enable designers to directly implement choice architectures (and principles such as asymmetric paternalism), allowing players to make firsthand decisions that influence in-game plots while building their sense of moral responsibility and self-efficacy [43]. In ridehail, interactive fiction (1) augments player understanding of driver labor while (2) revealing more growth-oriented and socially responsible choices.

	Replayability	Timed Challenge	Embedded Design			Ridehail-Specific		
			Obfuscating	Intermixing	Fictional Narrative	Ground truth answers	Playable in-ride	Driver-Passenger Interactions
 CrossRoads	✓		✓			✓	✓	✓
 Dilemmas @ Work	✓			✓				
 DRIVEN	✓		✓		✓		✓	
 TRIVIA RIDE	✓	✓		✓		✓	✓	✓
 DRIVING QUESTIONS	✓			✓			✓	✓
 TICKING ROADS	✓	✓	✓		✓		Mobile-Only	

Table 1. How game prototypes met game (including embedded) design and ridehail-specific criteria

3.3 Ridehail-specific Criteria

We conducted pilot interviews with 2 drivers (D1, D2 – recruited from past studies) and 3 passengers (P1-P3) to elicit ridehail-specific requirements to complement our own knowledge of app-based labor.

3.3.1 *In-Ride Compatibility.* Pilot passengers expressed a common preference for “*lightweight*” and easy-to-pickup games to minimize chances of car sickness, although P2 and P3 also desired realistic simulations of ridehail driving. Pilot driver D2 suggested using the Octopus tablet currently in their car to integrate content in more natural ways while D1 cautioned how embedded content should not come across as a way for drivers “*vent your complaint*” to passengers.

3.3.2 *Driver-Passenger Interactions.* When discussing preferred genres, P2 indicated interest in simulations that shed light on how drivers interact with and “*talk to the person[/rider] in the backseat*”, since they’re not a fan of actual driving whereas P3 suggested interactions where “*you have to talk to the driver, or engage with them*”.

3.3.3 *Ground Truth Answers.* To most effectively reconfigure and transform the role of passengers, concepts should convey accurate information regarding ridehail labor. Factual and apolitical content makes it easier for players to trust and learn, in addition to reaching larger audiences.

4 PLAYABLE PROTOTYPES OF PASSENGER-FACING IN-RIDE GAMES

Drivers of the first two workshops made several concrete and actionable suggestions that informed our decisions to remove two prototypes – **CrossRoads** and **Dilemmas @ Work**, which we describe more in the Appendix. Table 2 offers a timeline that summarizes workshop order, and the presence or development of prototypes between sessions. Early driver feedback also inspired the addition of **TICKING ROADS**, a driver-facing settings page to indicate (1) topics to avoid in discussion and (2) currently preferred interaction levels with passengers, which range from “*not at all*” to “*anytime*”, the selection of which dynamically reconfigures the passenger-facing menu screen – *e.g.*, more interactive games get grayed out when the driver chooses “*not at all*”.

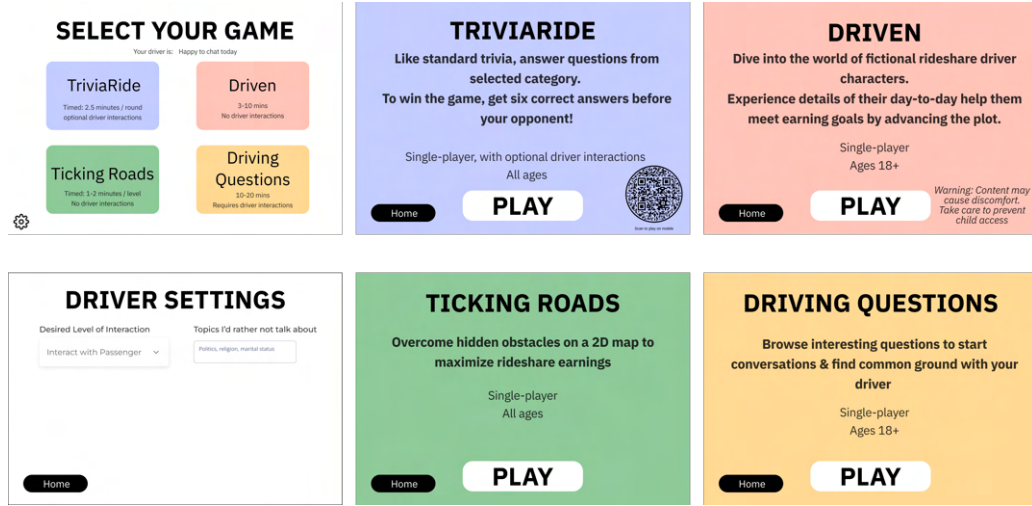


Fig. 1. Menu informs passengers about expected content and interactions, as well as driver preferences when selecting a game

4.1 Visual Novel: **DRIVEN**

DRIVEN is a visual novel (*i.e.* a digital narrative) with interactive decision points that branch into multiple storylines – creating spaces for players to experience ridehail conditions with psychological distance (§3.2.3). Pilot passenger P1 highlighted to us the value of more casual and broadly accessible experiences – *i.e.* “*games that are much more about the story ...messaging ...[to] make it easier, more accessible for everyone, even people who are not used to playing games.*” Consequently, our choose-your-own-adventure offers two driver characters whose plotlines unfold based on decisions that players make on their behalf to work towards income goals while balancing stressors from passengers and life.

While **DRIVEN** was heavily inspired by *Cherry Picker* [12] and *The Uber Game* [45], it uniquely adopts a first-person perspective that leverages the persuasive power of procedural rhetoric (§3.2.3) to situate players within more complex issues – *e.g.*, work-life balance, algorithmic management, deadheading, driver-passenger (power) dynamics and interactions. For instance, **DRIVEN** foregrounds driver responsibilities – *e.g.*, those arising from family or primary occupations (main jobs) – which prime players to consider the non-driving goals of drivers, which subtly intermixes narrative content while redirecting player attention away from the game’s persuasive intent – *i.e.* obfuscation. The divergent branching storylines provide high replayability, offering varied experiences across play sessions that drives

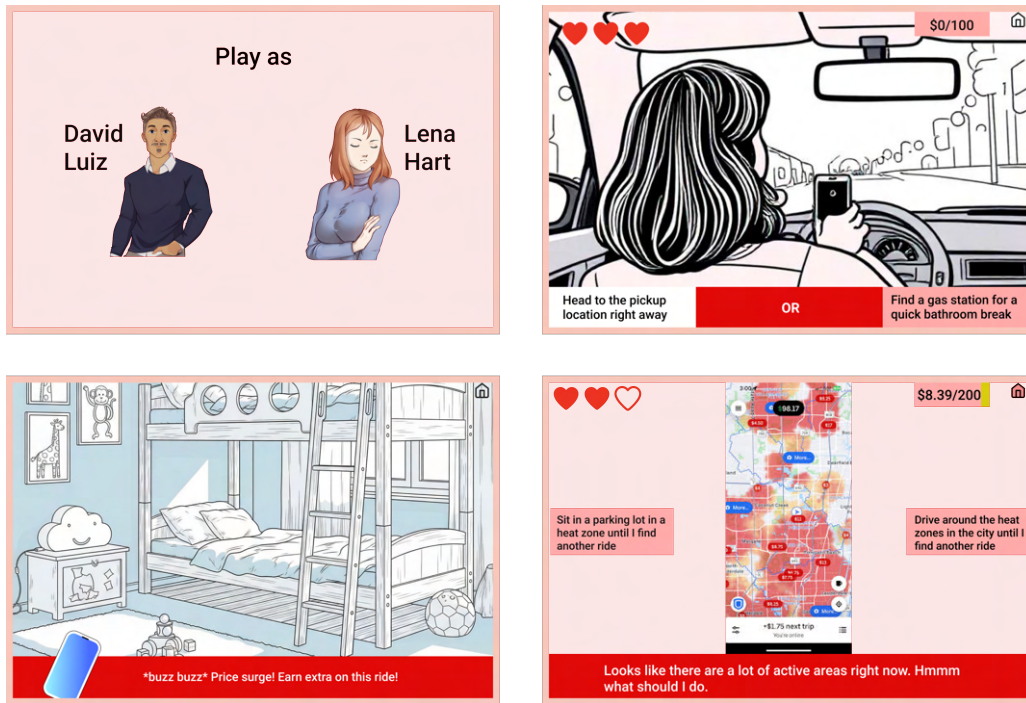


Fig. 2. **DRIVEN** is visual novel with point-and-click options that advance plotline of two NPC ridehail drivers

player engagement and transformed understandings of driver-centric perspectives. Peripherally, passengers in-ride gameplay can also prompt further questions, conversations or reflections with the driver.

4.2 ? TRIVIA RIDE

Based off of the multiplayer game *Trivia Crack*, **TRIVIA RIDE** challenges players to reach six correct answers before a hypothetical opponent. The game contained a mix of general trivia – spanning categories like Social Studies, Science, Pop Culture, and the Arts (e.g., ‘*What sport has been played on the moon?*’) – intermixed with ridehail-related policies or facts (e.g., ‘*What law classified drivers as independent contractors in CA?*’). All questions contain a verifiable ground-truth answer, and the ridehail questions are designed to require minimal background knowledge, allowing the players to quickly learn from prompts like ‘*Which location is the **most** lucrative for rideshare driving at 3am?*’.

TRIVIA RIDE also incentivizes driver-passenger interaction by giving “redemption” chances after incorrect responses, delivered through an interface prompt that encourages the player to verbally ask their driver for help. This design was motivated by D2.2’s suggestion of “*adding a driver-passenger collab mode would be super cool ...A game where we solve a puzzle together*”. In later versions of the game, the first-to-six objective was replaced with a timed, point-based system, following D2.2’s recommendation that “*putting a timer would be very good [to provide] urgency to answer the question*” – in alignment with the timed challenge criteria (§3.1.2).

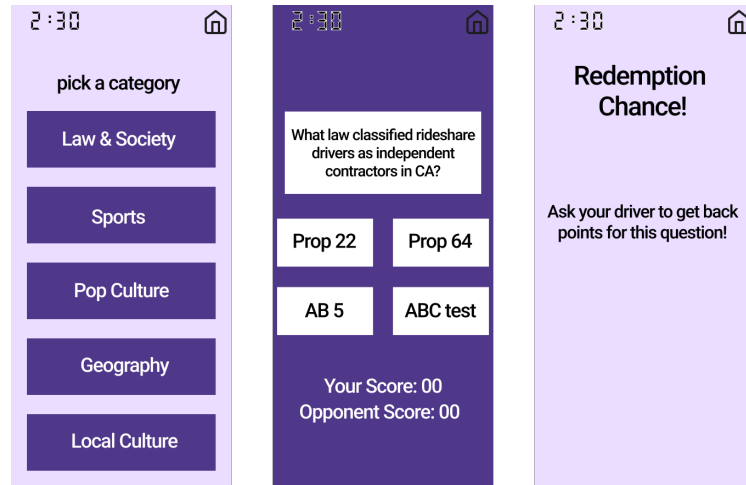


Fig. 3. **TRIVIA RIDE** is a timed challenge with optional driver interactions and embedded ridehail concepts

4.3 **DRIVING QUESTIONS**

Inspired by the game *We're Not Really Strangers* (WRNS) and driver D1.2's suggestion for more driver-passenger interactions and emotional connections, **DRIVING QUESTIONS** is a conversation prompting game that serves as boundary object to initiate and mediate conversations during a ride – helping keep discussions topically related to driving labor while allowing the driver-passenger pair to get to know each other as people. We repeatedly updated the content to minimize intrusive content (concepts embedded include mental health impact, logistics, take rate) while intermixing more locally-grounded questions. Since driver-passenger pairings are almost always unique, replayability of **DRIVING QUESTIONS** is strong and its selection of 18 conversational questions also increase response variation with every new driver-passenger combination.

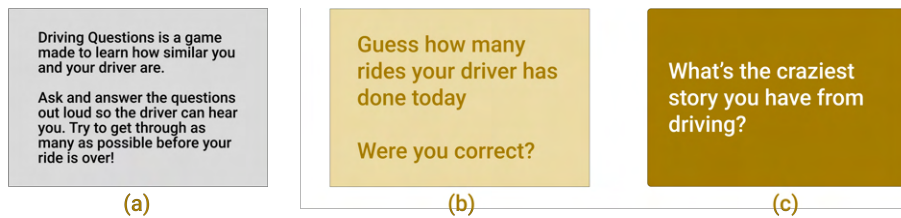


Fig. 4. **DRIVING QUESTIONS** bridges the driver-passenger social gap with conversation prompts for passengers (b) and drivers (c)

4.4 **TICKING ROADS**

Borne from the recommendation of drivers from the first session, **TICKING ROADS** is a time management game (similar to *Diner Dash* and *Overcooked*) that surfaces stressors of the road to players by simulating ridehail obstacles on a map. Players undertake the task of picking up passengers at designated locations and receive immediate feedback when performing actions (e.g., move around, speed up/down, wait at pick up location). By framing stressors as obstacles and introducing time pressure, **TICKING ROADS** diffuses (or obfuscates) the tensions of exposing dangerous road

conditions. It also creates psychological distance between the player and realities of driving on the road by presenting a fictional, abridged simulation of logistical burdens that drivers regularly encounter.

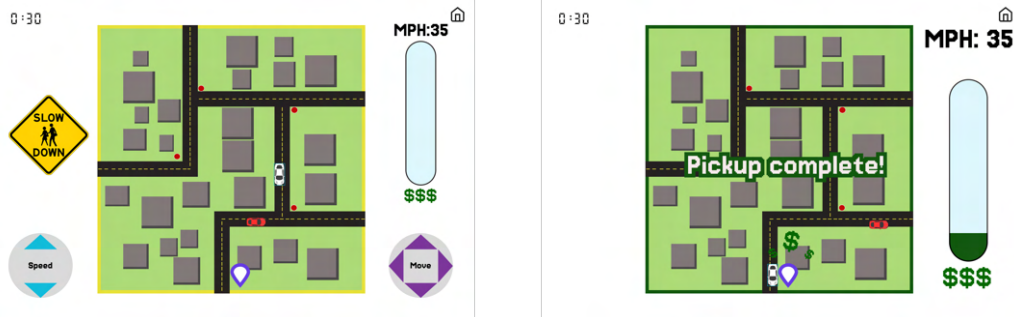


Fig. 5. **TICKING ROADS** simulates passenger pickup (logistics) and immediate feedback from “driving” a car on the map

5 METHODS

Given the unconventional and multi-disciplinary problem space (i.e. raise awareness around realistic ridehail driving conditions), intended audience (i.e. passengers) and goal (i.e. motivate and mobilize consumers to care and advocate for driver labor), we followed the (Tandem) Transformational Game Design process [32, 152]. All interviews

5.1 Phase 1: Goal Delineation

Starting with a cycle goal delineation, we synthesized relevant and recent **ridehail literature** to surface key but potentially underexposed conditions – summarized in §2.1. In parallel, we identified potential techniques and genres from scholarship on transformational, serious and persuasive **game design** that may support our goal of motivating passengers’ perception change around ridehail driving conditions (§3.1). Next, we conducted **formative interviews** with 2 drivers and 3 passengers (as mentioned in §3.3) to garner initial ideas and understanding around latent topics that drivers desire to communicate to passengers, levels of comfort and concern for a passenger-facing game addressing such issues, as well as preliminary reactions around (and suggestions for) potential game genres.

5.2 Phase 2: Iterative Game Prototyping

After the initial round of goal delineation, we presented three prototypes to drivers across two workshops, followed by a second round of goal refinement before moving on to passenger feedback. Table 2 shows when prototypes were conceived, developed, presented and eliminated while Table 5 shows how people perceived prototypes across sessions.

5.2.1 Driver Workshops. Based on feedback from the formative study in Phase 1, we began implementing three game prototypes (§4) and presented these to drivers across DW1 and DW2. In driver sessions, we inquired about issues they prioritized to share with passengers, probed for initial reactions and hesitation to prototypes and embedded concepts, and elicited ideas for alternative game designs and concepts to embed that align with the overarching goal – see supplementary materials for detailed driver workshop protocols. Then, we presented three initial prototypes – **TICKING ROADS** was not yet conceived/developed – through two initial workshops with drivers (DW1, DW2). Afterwards, we completed another round of goal delineation [152] to

- (1) conceptually map (see supplementary materials) relevant concepts in ridehail driving to refine §2.1

- (2) highlight concepts at the intersection of
- what drivers prioritize to communicate with passengers and
 - conditions and vulnerabilities that are underexposed to passengers.

Incorporating insights from goal refinement and initial driver feedback, we iterated on game mechanisms and content, and implemented **TICKING ROADS**. To continuously adapt prototypes based on feedback from both stakeholder groups, we held three more driver workshops (DW3-DW5), interspersed among passenger sessions.

	DW1	DW2	DW3	PW1	PW2	DW4	PW3	PW4	DW5
CrossRoads	✓					✗			
Dilemmas @ Work	✓					✗			
DRIVEN					✓				
TRIVIA RIDE					✓				
DRIVING QUESTIONS					✓				
TICKING ROADS						✓			
<i>*Menu</i>							✓		

✓ = Presented ✗ = Eliminated  = Idea Conception  = Under Development

Table 2. Timeline of prototypes presented across workshops. DW1-5 were driver-facing workshops while PW1-4 (in gray) represent passenger-facing ones. Progression of the menu is also included (*).

5.2.2 Passenger Workshops. Next, we probed passengers to gather initial understanding and concerns around ridehail driving, evaluations of prototypes, and hesitations or ideas for alternative interactions that align with our study goal. Each session started with “Character Card” introductions where passengers took turns sharing their name/location/experiences on sticky notes of specific colors, which we used to collect responses to questionnaire-style prompts about their (1) knowledge of ridehail labor (2) level of empathy with drivers (based on questions adapted from the QCAE [125] and IRI [80]) and (3) willingness advocate for drivers. Then, each passenger took turns playing a game of their choice (presented via Figma prototypes) and rated their interaction experiences along seven dimensions: fun, replayability, sneakiness (at embedding ridehail concepts), ride-friendliness, lightweight vs taxing, recommendability and how thought-provoking it was. At the end of each session, we re-administered the questionnaire to survey perception

Table 3. Passenger demographics across four workshops (labeled PW1-4), with individual passengers labeled “PW1.1” etc.

	ID	Age	State	Gender	Income	Platforms	Drives to Commute	Employment
PW1	P1.1	18-29	MA	NB	\$12k - \$50k	Uber (4.94), Lyft	<Once a year	5-10 years
	P1.2	18-29	PA	M	\$12k - \$50k	Uber (4.81)	Never	<3 years
	P1.3	18-29	CA	M	\$100k - \$200k	Uber, Lyft (5)	Everyday	<3 years
	P1.4	18-29	TN	M	\$12k - \$50k	Uber, Lyft (5)	A few times a year	3-5 years
PW2	P2.1	18-29	CT	M	\$50k - \$100k	Uber (5)	Everyday	3-5 years
	P2.2	65+	NV	F	\$50k - \$100k	Uber (5)	Never	10+ years
	P2.3	18-29	IL	NA	\$12k - \$50k	Uber: (4.68)	Never	3-5 years
	P2.4	30-44	NY	M	\$100k - \$200k	Uber (4.9), Lyft	Never	10+ years
PW3	P3.1	18-29	NJ	F	\$50k - \$100k	Uber (4.98), Lyft (5)	Few times / year	<3 years
	P3.2	18-29	MA	M	\$0 - \$12k	Lyft (5)	Everyday	3-5 years
	P3.3	45-64	TX	F	\$50k - \$100k	Uber, Lyft (5)	<Once a year	10+ years
	P3.4	30-44	PA	M	\$12k - \$50k	Uber (4.9), Lyft (4.9)	Never	10+ years
PW4	P4.1	18-29	MA	M	\$50k - \$100k	Uber (5)	Few times / year	<3 years
	P4.2	18-29	NJ	F	\$12k - \$50k	Uber (4.99)	Never	3-5 years
	P4.3	45-64	FL	M	\$100k - \$200k	Uber (4.90) Lyft (5.0)	>Once a week	10+ years

changes around, knowledge of, empathy with and motivation to advocate for ridehail labor, which serve as proxy measures to help us gain initial understanding of whether game design can transform users to engage in more solidarity with drivers and socially conscious consumption behaviors. Supplementary materials contain full ranking results and detailed workshop protocols.

	ID	Age	City	Gender	Education	Status	Ridehail Income	Hrs/week	Ridehail Experience
DW1	1.1	45-54	PA	M	High School	FT	Essential for basic needs	N/A	6 years
	1.2	45-54	IL	M	High School	FT	Essential for basic needs	N/A	4 years
	1.3	18-29	CA	M	Bachelor's	FT	Essential for basic needs	25-40	3 years
DW2	2.1	45-64	NC	F	Some college	FT	Essential for basic needs	25-40	One month
	2.2	30-44	NY	M	Associate's	PT	Nice but not essential	10-25	2 years
	2.3	18-29	GA	F	Bachelor's	PT	Essential for basic needs	25-40	3 years
DW3	3.1	30-44	TX	M	Bachelor's	PT	Essential for basic needs	25-40	5 years
	3.2	30-44	WA	M	Bachelor's	FT	Essential for basic needs	40+	6 years
	3.3	18-29	NY	NB	Associate's	FT	Essential for basic needs	25-40	3 years
	3.4	18-29	NY	F	Post-Graduate	N/A	N/A	N/A	2 years
DW4	4.1	30-44	CA	N/A	Associate's	PT	Nice but not essential	25-40	5 years
	4.2	30-44	FL	M	Associate's	FT	Nice but not essential	25-40	7 years
	4.3	30-44	CO	NB	Bachelor's	PT	Nice but not essential	25-40	4 years
	4.4	18-29	TX	NB	Associate's	FT	Nice but not essential	25-40	3 years
DW5	5.1	30-44	PA	M	Bachelor's	PT	Essential for basic needs	10-25	7.5 years
	5.2	30-44	IL	M	Bachelor's	FT	Essential for basic needs	25-40	5 years
	5.3	18-29	PA	NB	Bachelor's	FT	Essential for basic needs	25-40	8 years
	5.4	45-54	PA	M	Some college	N/A	Nice but not essential	N/A	2.75 years
	5.5	45-64	GA	M	Bachelor's	FT	Essential for basic needs	25-40	6 years

Table 4. Driver demographics across five workshops (DW1-5), individual workers are labeled “DW1.1” etc.

5.3 Recruitment

During formative interviews, we recruited 2 drivers (D1, D2) based on contacts from prior studies, as well as 3 passengers (P1 – P3) based on convenience sampling from our home universities. For workshops, we reached out to (1) drivers from past studies, (2) subreddits, Craigslist posts and physical flyers in local professional communities. All formative interviews and workshops were conducted via Zoom. Both participants groups were compensated at a rate of \$60/hour and selected based on eligibility, location and experience levels, indicated by pre-study screening forms. Tables 3 and 4 summarize passenger and driver demographics ¹.

5.4 Thematic Analysis

After our workshops, three researchers open coded all 12 hours of workshop transcripts (transcribed by Otter.ai) to extract relevant themes and opinions on improvements for each prototype. Then, we combined all individual driver codes in an affinity diagram to map out common ideas, extracting 8 main categories: existing practices/strategies, frustrations, reactions to prototypes, current consumer perceptions, underexposed knowledge, design objectives and alternative gamified interactions/interventions. The first five categories helped us understand how well the prototypes capture realistic driver experiences, while the remainder guided our next iterations of prototype adjustments – these codes also helped eliminate less effective prototypes and introduce new features. In particular, codes in the category of

¹Due to an oversight to record the meeting, participant data from DW4 are based primarily on internal notes

reactions to prototypes informed the construction of Table 5. We considered a prototype “preferred” in a workshop if more than half of participants expressed affinity to it. If a session was divided on their sentiments – roughly half positive and half negative – we categorized it as “neutral”. Remaining prototypes that received mainly negative reactions were considered “resisted”. We halted recruitment and data collection when new themes no longer emerged.

5.5 Positionality

Our study activities (from recruitment, consent and data collection) followed procedures of an IRB-approved protocol. However, working with sensitive topics across two stakeholder groups carries unique dangers that our internal team continuously deliberate on to reach further understanding of our relative positions [146]. We reflect on ways to center driver experiences and reduce replacing their voices and opinions with our own values and epistemologies, paying particular attention discussing vulnerabilities in ways that uplift and empower, rather than silence, suppress or overshadow worker experiences. Towards passengers, we disclosed our intent to raise awareness around ridehail driving conditions at the start of each session. Yet we still acknowledge that even upfront notices cannot fully eliminate risks of imposing our own values and positions onto passenger participants, which may cause discomfort – especially when using techniques such as obfuscation. Thus we also explicitly reminded participants about their right to opt out at any time throughout the session. Our team members receive training in Computer Science, Media Arts & Sciences, Software Engineering and Human Computer Interaction, where two authors have experience researching and working with ridehail drivers. One author has part-time experience working for a food delivery platform, while two authors have extensive experience laboring across service occupations.

5.6 Limitations

While our *Character Card* activity during passenger workshops aimed capture early evidence of how passengers perspectives changed before and after experiencing gamified interactions, these results are limited by the size of our participant sample. Future works might consider more scalable evaluations of similar tools to more rigorously examine the impact of such interventions on player understanding and advocacy for driving conditions. Our US-based investigation can also limit external validity and generalizability – both geographically and culturally. For instance, how do more collectivist (e.g., Asian-Pacific) societies or nations in the Global South (e.g., Chile [3]) receive or resist such mechanisms of control? How would a post-colonial lens of analysis offer different results between migrant workers of the US vs Europe [102]? We hope this study offer a case study that might inspire compare analysis with other national, cultural and geographic contexts.

6 RESULTS

Below we first report how passengers received the game prototypes, early evidence of how they shifted their perceptions of ridehail, and their knowledge gaps that the probes revealed. Next, we discuss driver preferences for purposeful and personalized prototypes that initiate unforced and fulfilling interactions with passengers, as well as driver perspectives on and experiences with common passenger knowledge gaps. Finally, we share how both stakeholder groups reacted to proposed driver-passenger interactions, including reasons for hesitations and ways to better motivate engagement.

6.1 Passenger Receptions to Gamified Interventions, Early Perceptual Changes & Knowledge Gaps

During formative interviews, all passengers reflected on gaps in their understanding of ridehail work, noting that incorporating content related to ridehail conditions would motivate engagement with the game. For instance, P1 related

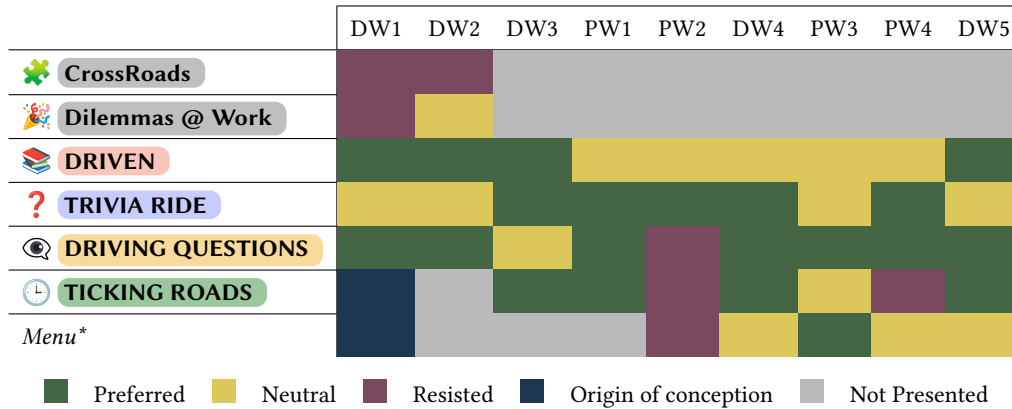


Table 5. Heatmap shows participant perceptions of prototypes across workshops, which are ordered chronologically. DW1-5 represent driver-facing workshops and PW1-4 shows passenger-facing ones. Feedback for the menu selection is also included (*).

how “[he]’ll be more inclined to try it out” while P3 shared that “[she] would definitely play a rideshare driver simulator ...where your goal is to get from one place to another”. Building on such potential, we next report how passengers received our game prototypes during workshop sessions (§6.1.1), early evidence of how the play experience changed their perceptions towards ridehail conditions (§6.1.2), and prominent knowledge gaps that the game probes revealed (§6.1.3).

6.1.1 Passenger Feedback on In-Ride Game Prototypes. We overview how drivers ranked prototypes along (1) playability heuristics and (2) their abilities to provoke reflective thinking around ridehail conditions.

Fun vs Provocation, Replayability & Recommendability. Passengers found **DRIVING QUESTIONS** the most fun (PW3, PW4) and replayable (PW1, PW3, PW4). **TRIVIA RIDE** came in second for fun, preferred by passengers like P2.1, who found it “a little bit fun — the interface was very simple, and so that made it, not super thrilling. But I enjoy trivia ...and I really like the idea of having questions that are specific to the area where you’re riding, I found that very inventive”.

In contrast, **DRIVEN** was rated as least fun but most provocative (PW1, PW3, PW4) and second most replayable. Largely due to technical issues, **TICKING ROADS** consistently placed last for replayability – P4.3 noted how “controlling issues were a pain, it was kind of wonky”, surfacing a tension between immersion and implementation effort. Initially, PW1 ranked **TICKING ROADS** as least recommendable. After we implemented improvements based on initial feedback from PW1, passengers in PW2-4 ranked it as most recommendable.

Ridehail-specific Rankings. Regarding ridehail content, we asked passengers to rank each prototype by how (1) “sneaky” it was at embedding driving-related content (a proxy measure for success of obfuscation) (2) ride-friendly it was and (3) lightweight or taxing it was to navigate game mechanisms and content. **TICKING ROADS** was most successful at obfuscation, with P3.4 admitting how “that was pretty sneaky. I’m not gonna lie. I didn’t even realize it. Thought I was [just] driving around” while **DRIVEN** was most ineffective at obfuscation. The conversation-prompting prototype **DRIVING QUESTIONS** was determined to be most ride-friendly, while **TRIVIA RIDE** came in close second. The lightweight-taxing ranking measurement was interpreted by passengers along both dimensions of gameplay mechanics and content: **DRIVEN** was deemed most emotionally-taxing since it is “frighteningly realistic ...and depressing, and for once it seems like accurate for a lot of people’s [driver’s] situations” (P2.4). By contrast, **TRIVIA RIDE** was considered most lightweight, both in terms of content and mechanics, although its ridehail content was most exposed, obvious and not well obfuscated: “they were very apparent” - P2.3.

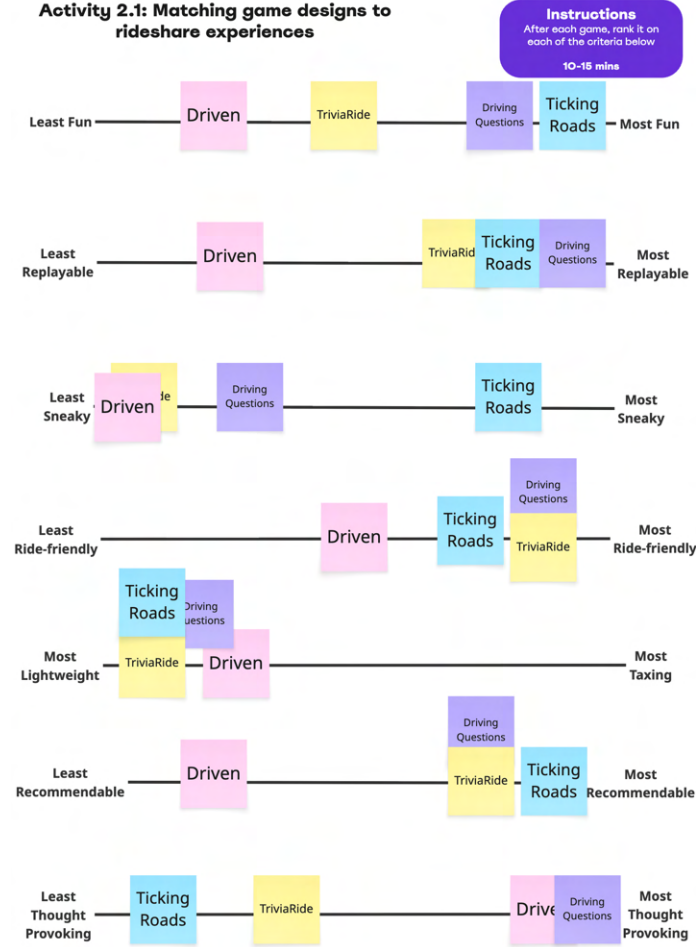


Fig. 6. After playthroughs, passengers ranked each prototype along game heuristics, an embedded design criterion (“sneakiness”) and ridehail-specific criteria (ride-friendliness, ability to provoke) – Figure shows example from PW1.

6.1.2 How Games Evolved Passenger Perceptions & Solidarity with Ridehail Drivers. Leveraging the *Character Cards* (§5.2.2) activity, we surveyed how passengers perceived, empathized with, and desired to advocate for driving conditions both before and after introducing prototypes. Supplementary materials show how passengers placed themselves on these scales across all four workshops. Below, we report changes in their (1) knowledge around ridehail conditions as well as (2) empathy with and willingness to advocate for drivers – which we consider as proxy measures for solidarity.

Shifts in Passenger Knowledge of Ridehail Driving. When probed about their readiness to drive for ridehail themselves (i.e. *How prepared do you feel to be a full time rideshare driver?*), confidence increased slightly in PW1 and PW4, whereas all passengers in PW2 felt unprepared before and after. Estimations of driver take-home rate was unanimously lowered across all four workshops, suggesting that prototypes helped passengers reach **more accurate understandings of platform take rates**. Meanwhile, passenger estimations of deactivation rates surged – especially in PW3, where ratings

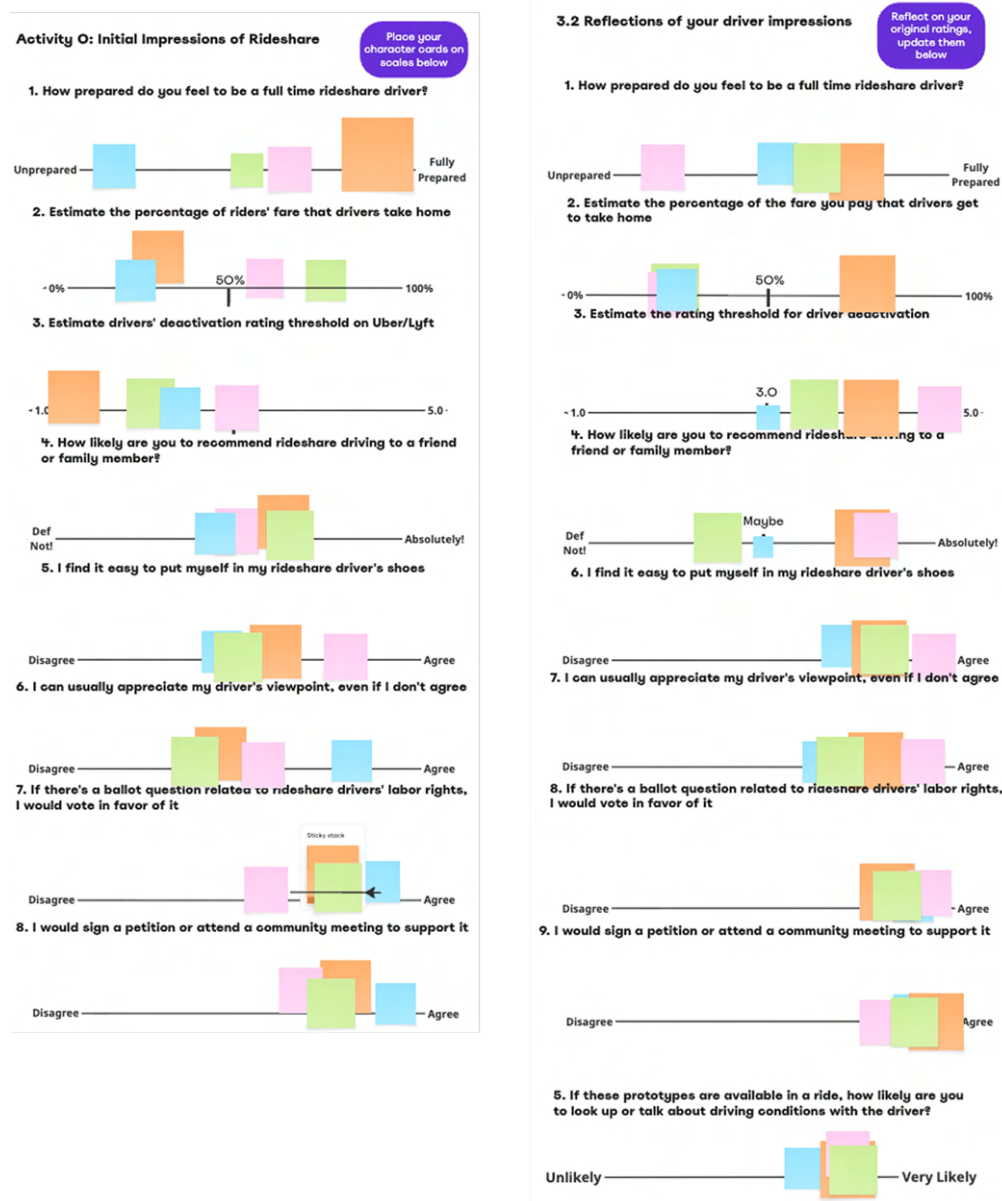


Fig. 7. *Character Cards* (with each color representing a passenger) helped us survey passenger understanding, empathy and willingness to advocate before and after each workshop. Example above show (changes in) responses from PW3.

flipped entirely – see Figure 7 – indicating a significant knowledge gap around how heavily ratings impacted the job stability of drivers. Passenger willingness to recommend ridehail driving remained largely unchanged.

Early Indicators of Change in Passenger Empathy, Solidary & Advocacy for Ridehail Labor. Empathy-related questions (i.e., “I find it easy to put myself in my rideshare driver’s shoes”; “I can usually appreciate my driver’s viewpoint, even if I

don't agree") were introduced after the first passenger workshop (i.e. PW2–PW4). Across these sessions, passengers showed slightly higher sensitivity to driver perspectives, with effects most prominent in PW3. In the later three workshops, we also added two items to probe passenger willingness to advocate for ridehail drivers. Across all sessions, **passengers became more motivated to vote in favor of "a ballot question related to rideshare drivers' labor rights"** with those in PW2 and PW3 expressing absolute agreement by the end of their sessions. In response to the follow-up question (whether they would "sign a petition or attend a community meeting to support it") all passengers showed slightly higher inclinations to advocate.

6.1.3 Passenger Knowledge Gaps Surfaced by Game Prototypes. Based on how passenger reacted to embedded ridehail content, we next present prominent knowledge gaps that our prototypes helped surface. For instance, P3.1 "feel uncomfortable asking them how much they're getting paid, or their take-home pay", let alone more personal inquiries about what drivers miss (§4.3), since "it can be painful when you leave your home country" (P2.2). However, the optional yet inviting nature of games like **DRIVING QUESTIONS** serve as boundary objects for mediating such conversations, even among more quiet passengers – P3.3 identified as "an introvert [...] and I have trouble starting conversations [...] this game would help me be more talkative, because it gives me things to say".

Pay rates. All passengers of PW1 & PW3 were "surprised about the 30% [take-home rate our drivers estimated.] I didn't realize that it could be that low. That's bad." P2.2 (a more sympathetic passenger who regularly asked drivers about rising take rates) observed how "all the drivers are anxious to answer [...] to ensure] riders know that even though [passengers] are paying more, [drivers] are not making more". **DRIVEN** was most effective at bringing out reactions towards low pay rates, leaving P3.2 outraged: driving "80 trips for \$65 is crazy!" **TRIVIA RIDE** was also clear at communicating low pay rates, with P2.3 and P3.1 noticing a question about most profitable hours of operation. While **DRIVING QUESTIONS** carried potential to spark driver-passenger discussion around pay (P1.2), many passengers hesitated to discuss the financial and potentially sensitive topic – see §???. Although **TICKING ROADS** was designed to show how obstacles influence a driver's earnings, players generally did not notice these financial effects – indicating the concept may need to be presented more clearly or reinforced through replays.

Pickup Logistics. The time that drivers spent waiting for passengers during pickup is a commonly overlooked factor that affects driver pay. P3.3 exhibited concern about how their location may impact driver earnings "I live in a suburb ...and ...I don't think they're getting paid between the different ones. ...I always wonder if they're making a lot less money and just driving basically for free between the places." The time pressure in **TICKING ROADS** guided passengers like P4.3 to empathize about wait times at pick up: "that must be frustrating for the the driver to wait for the person to come up." Similarly, P4.2 noticed through the plot of **DRIVEN** how drivers must decide on an exact pickup location when directions from the app are not clear: "when drivers are going to somewhere that has an entrance and an exit, and knowing where to pick up someone." With **DRIVING QUESTIONS**, P2.2 was the only one who noticed a question centering wait times during pickups, even though all players went through the same set of questions. Given its need of ground-truth answers, we struggled to embed factual content on pick up logistics within **TRIVIA RIDE**.

Rating Pressures & Passenger Expectations of Service. In terms of mechanisms, **TICKING ROADS** was most effective at eliciting passenger reactions to behaviors of impatient riders, with P2.3 rating the prototype as "provoking, because you have a person texting you angrily when you're trying your best ...it'll teach them to put themselves in the drivers' shoes." Similar to frustrations that drivers often feel about road conditions, P4.3 also grew annoyed at the simulated passenger in the time management game who was "Texting, 'where are you?' But there's obviously a car right there, lots

of traffic”. **DRIVING QUESTIONS** reminded P1.2 of “one driver ...he was telling me about ...all the passengers that he picked up during the night, which are usually all like drunk college kids”, suggesting that the conversational nature of the interaction can help drivers more comfortably share experiences of poor passenger behaviors. Despite how **DRIVEN** embedded consequences of receiving bad ratings as a result of interactions with passenger, the prototype did not actually demonstrate tangible consequences (e.g., deactivation, reduced work availability) beyond small pay differences — which may have caused passengers to not notice the embedded concept. **TRIVIA RIDE** also did not embed concepts on rating pressures.

Platformic Management & Long-term Consequences. While passengers may rarely observe the longer-term tolls and costs that ridehail cause for drivers, psychologically-distanced narrative elements in **DRIVEN** (e.g., pop-up app notifications at home) helped surface a key tension that last over time — family-life balance — which was noticed by every passenger who played through (P1.4, P2.4, P3.3, P4.2). In their second playthrough, P2.4 started to **anticipate extractive platform tactics**: “I feel like they’re gonna ping me and say, ‘Why are you taking so long use the bathroom?’ ” The modular and timed nature of **DRIVEN** again made it unsuitable for addressing psychological control, even though D1.2 saw potential for simulations like **TICKING ROADS** to give character to “the big evil ride sharing company”. While the conversational nature of **DRIVING QUESTIONS** could prompt discussion around long term consequences and control, passenger discomfort with potential intrusion may hinder in-depth probing (§??).

6.2 Towards Driver-Centered & Integrated In-ride Gaming Interactions

Next, we overview driver receptions of game prototypes, including their preferences for immersive and personalized environments that help passengers feel more at ease and willing to engage in interactions with the driver. Then, we describe how drivers observed passenger knowledge gaps, including **covert logistical, emotional and immaterial labor** that range from general car maintenance (e.g., gas, oil changes) to in-ride services — e.g., keeping comfortable temperature and mood while ensuring safety.

6.2.1 Drivers Prefer Memorable, Contextualized but Unobtrusive Games. Drivers expressed strong enthusiasm for the potential of games to create **purposeful, immersive and memorable** gamified experiences around ridehail knowledge. For instance, D2.2 emphasized how “the story game: it actually rocks — taps into emotions, memories and creativity, and then it’s relatable and often openly hilarious” and imagined cases of “making [passengers] go ‘yeah, man, I never thought about it like that’ ”, causing many (D1.1, D2.1, D2.2, D2.3, D3.4) to rank **DRIVEN** highest: “love the story games because [...] it builds empathy in a more subtle way” (D3.4).

An immersive narrative experience also help drivers create an environment that **distracts passengers from time pressures** of reaching their destination. D3.2 imagines how “When a rider is into a story game ...the whole mood in the car becomes more quiet, more relaxed, no pressure, no rush. They’re just absorbing something soft and engaging.” But alongside immersion, D2.2 also cautioned against cognitively demanding interfaces: “style choices that spark a bit of fun without demanding too much attention”, while D3.1 warned about the “risk of getting halfway through something, and then, boom, you are there [the ride] stop[s], and it actually ends awkwardly. So for me, the sweet spot would be not keeping the stories short, flexible and rider-led”. Through this relaxed and non-confrontational approach, drivers shared desires to communicate latent knowledge including pickup logistics (D1.2, D3.1, D3.2), long hours (D2.1, D2.3), unreasonable passenger expectations (D2.1) and behaviors (D2.3, D1.3), traffic (D1.3, DW4) — some of which drivers would’ve been difficult to broach since they “don’t want to make [passengers] scared to be in the ride with you” (D1.1).

Drivers who have experience with Octopus tablets noted that its existing trivia content was not designed for ridehail contexts, leading them to prefer alternatives like **TRIVIA RIDE** (ranked highest by D3.1, D3.3, D2.3) that feel more **personalized, local** and **mood-aware**. For instance, D2.2 “*want games that feel like they belong in the rideshare world, not like they were copied from somewhere else and shoved into my car*”, suggesting instead interactions with “*more personalization, more mood awareness, more empathy, fun*”. Numerous participants (D2.2, D3.1, P2.1, P2.4, DW4 members) also suggested location-specific content, to “*add a local city flavor, trivia about Seattle [...] we have landmarks nearby, or [...] which coffee shop this quote is from – it makes the ride feel connected to where we are, and breaks the ice faster than [games on] a plane*” (D2.2).

With a lighter interaction like **DRIVING QUESTIONS**, drivers saw potential to start off **unforced conversations** around topics of mutual interest. For instance, D3.2 discusses how this prototype “*changes the mood ...spark[s] some light interaction [between drivers and passengers and] also between the people onboard*”, in addition to engaging both quiet and extroverted passengers:

It’s a game, so people are not forced to to play it. So this is just one way in which you will create the conversation. If the rider feels like he/she is not compatible to play the game, then it’s okay. But if one is ready for that, then absolutely, there are some people who are extroverted ...and I think this would match quite well with my riders. -(D2.2)

For drivers like D5.5, such connection with (introverted) passengers generates an **intrinsic sense of fulfillment**: “*I really love seeing where they say that they prefer to be quiet, and you could still get a conversation out of them – makes me feel like I’m doing my job, plus more*” while simultaneously helping passengers “*keeps them off their mind of traffic*”.

Since **TICKING ROADS** was conceived at workshop DW1, it was only presented in DW3-5 (see Table 5), but it was a favorite for drivers like D5.2, who found it relatively “*more interactive*.” Drivers in DW4 also saw its mechanics and graphics as helpful simulations of driver experiences, especially for populations under (time pressure) such as busy parents managing small children onboard. However, D2.2 stresses that the interactions should engage passengers without disrupting their ability to focus on driving: “*as someone who is behind the wheel all day, I’m gonna think about mood, safety and how natural it all feels. If a game fits seamlessly into the ride, awesome. If not, then it just adds noise, pressure or openness*.” In general, drivers saw potential for **TICKING ROADS** to quickly capture attention, which **ease passenger anxieties and pressures**, with D3.2 observing how it can “*reduce rider frustration during delays [and] ...construction. Time[d] games give the rider something else to think about, so it lowers complaints and makes the ride feel more like a break than a hassle*.”

6.2.2 Driver Reflections on Latent Labor Conditions. Across workshops, drivers leveraged game prototypes to consider and reflect on experiences and labor that typically remain concealed from, or gets overlooked by passengers. While these are ordered by drivers’ expressed priorities, we note how they also **show more surface-level, salient and cognitively-obvious symptoms of algorithmic control first, followed by more nuanced, subtle and psychological reasons that cause harmful conditions**, demonstrating the ability of game probes to elicit more underlying fears and rationales. In their experiences talking to passengers, pilot drivers like D1 shared how “*Very few [passengers] – maybe one or two – out of the couple thousands of rides I’ve done have asked me what my pay for that ride versus what they were paying. So I think [they] probably don’t know [or] don’t care*”. D2 similarly thought that only folks who worked on “*a gig app ...or if there’s somebody in their immediate circle of life (friends or family) that does it*” are likely to know anything about it, suggesting that passengers lack motivation, spaces and occasions to learn about latent ridehail conditions.

Pay Rates. Drivers face a wide variety of stressors in daily operations that plague their well-being, but consistently prioritized pay as the most important misconception to clarify for passengers. For instance, D3.2 shared how they would “love [for passengers] to know that I don’t really get the full fare they are paying. [...] Uber, first of all, take their cut, then I cover fuel, car maintenance, time – all these are swallowed by [what’s broadly considered] service.” A direct result of this gap is lack of motivation to tip: “[passengers] assume a lot of times that they don’t have to tip [...] they feel like their charge all goes to the driver” (D2.1). In sum, D1.1 explains **how pay dominates all other concerns** since adequate compensation can alter drivers perceptions of all other stressors:

“If we are investing our energy, our time, our efforts, frankly, we just don’t [want] a feeling of being taken advantage of [...] of being manipulated [...] of being not cared about. I want to feel like somebody gives a [expletive] about me for the energy I’m putting in, [like] I’m making the money I’m making.” – (D1.1)

As a result of low pay (“they take these risks simply because they’re not being paid”), D1.1 further describe how some drivers protest by “flip[ping passengers] to cash rides”², even though doing carries serious insurance and deactivation risks, as well as “a mandatory impounding of the[ir] vehicle [for] the charge of providing commercial transportation in a non-commercial vehicle” (D1.2).

Pickup Logistics. Drivers such as D1.3 detail how **platform mechanisms fall short to ensure timely passenger arrival during pick up**: “the customer doesn’t really pay that much more in wait time [...] it’s not enough to be prohibitive [...] It’s almost like [...] [platforms] want to encourage that”. Consequently, D3.3 explained how pickup delays lead to wasted gas and lost time that drivers could have used for additional trips: “the waiting time is really expensive, especially when gas prices go up [...] it could just eat into our profits [so bad], and I feel like only us drivers really understand that.”

By contrast, D1.2 described their prior experience working for multiple cab companies – where dispatchers would tell drivers to “get out of there” and “get another [passenger]” if the current one doesn’t arrive within “literally, in 30 seconds”. Meanwhile, various road conditions (e.g., parking availability) can delay pickup when driving for ridehail:

“I had to pick up someone, it was center of the city, and there’s literally is no parking, it’s red lanes on both sides – bus only lanes. You can’t drive in those lanes. You can’t stop, you can’t park [...] So I was sitting in that lane waiting [and] of course, the bus came through. I got a ticket.” – (D1.2)

Rating Pressures Coerce Unpaid Services & Labor. Another significant factor exerting psychological control in ridehail labor are passenger ratings of drivers, since strict thresholds can put many drivers at risk of deactivation by platforms [131] – making “ratings feel personal” (D3.1) and thus coerced into performing a variety of **unpaid labor in service of ratings**. Inside of a ride, drivers frequently described how passengers expect “a premium vibe” (D3.4), “control over traffic” (D3.1), whereas drivers grapple with a variety of unpredictable factors, stressors and costs. Outside the car, D3.1 also point out external factors: “construction, detours, sixth street chaos and airport gridlock”.

“Ratings can drop for stuff I can’t control: sometimes I’m reading the vibe and giving folks their space, not just ignoring them. We are not just driving, we are juggling through navigation, personalities and keeping things safe. The app itself is not perfect, it doesn’t show everything we are dealing with [...] I’m paying attention to your body language, your tone, your energy, from the moment you get [in] [...] If you’re if you’re chatty, I’ll match that. If you’re quiet or stressed. I’ll try to keep things calm and give you space so it’s little things – e.g., adjusting the music so it fits your mood, making sure the temperature is comfortable, choosing the smoothest route so you’re not bouncing around in the back.” – (D3.1)

²strategy where drivers and passengers mutually agree to pay in cash to avoid platform fees

Long-term Consequences of Algorithmic Control. Corroborating related work on psychological and algorithmic control, D1.2 described how platforms intimidate drivers with warning notifications (e.g., “You should only use last ride when you’re when on your last ride”) to discourage use of the feature since it would make driver supply less predictable. Unfortunately, D1.1 also discuss how many financially vulnerable drivers who primarily depend on ridehail earnings for a living cannot afford to challenge such tactics: “most drivers they don’t see the s***, they don’t understand it ...a lot of foreign nationals ...[because] when you’re here just trying to put food on the plate for your family, when you see a warning like that come up, you’re not going to f*** around with your only source of income”.

Beyond app mechanisms, **platforms also incentivize drivers to take short-term risks for small rewards** (e.g., bonuses or pay boosts) without regard for longer-term consequences. For instance, D3.3 described how drivers sometimes sacrifice bathroom breaks to maximize time toward bonuses:

Those few seconds, even though I pee fast, it can make the difference between capturing it and not capturing [the ride]. We shouldn’t be facing those choices. Most people, it’s within reason. They can go to the bathroom on their job without facing, bonus losses [...] The pay boost isn’t always worth the effort. [...] What if for that I have to go to a doctor, having to pay more money than I would’ve even got with a little bonus?”

6.3 Designing Interactions that Motivate Driver-Passenger Solidarity

To motivate and approach healthier interactions between passengers and drivers, participants shared several barriers, tradeoffs and incentives that can inform future designs of in-ride engagements between the two stakeholder groups.

6.3.1 Respectfully Navigating Driver-Passenger Social Boundaries. While **DRIVING QUESTIONS** presents an opportunity for further social interactions between drivers and passengers, both stakeholders expressed reservations to initiate conversation, surfacing how interactions trade off with safety.

Driver Vulnerability to Harassment. Reflecting recent reports of harassment problems for both drivers and passengers [142], participants such as D2.2 expressed concern around suggestive topics: “sexually suggestive or flirty questions? No, no, not, not the place [...] It’s gross, it’s unsafe, and it creates a very dangerous precedent, especially for for women drivers and riders.” Even male-identifying drivers like D5.2 shared common experiences of passenger-imposed “sexual harassment. I’ve gotten that a lot. Like: ‘hey, I’ll give you this money and you do this to me, or let me do this to you.’” In fact, D5.2 has even been rematched to a perpetrating passenger: “I’ve reported several times to the Uber people about behaviors, and once I got this offender [that I once reported] again as my rider – we were matched again”

Safe & Socially Appropriate Spaces. Up front, drivers feared to cause **social awkwardness** or instigate potential **compromises to safety**: “not every rider wants to interact. And if the game feels too personal or like a distraction from driving, it can backfire” (D3.1). Safety resonated with several participants (P1.1-3, D2.2, D3.2-4,) – D1.2 for instance, expressed approval with the varied interaction modes available in **DRIVING QUESTIONS** because “it’s safe[r] when [the passenger] talk versus the driver [...] We don’t want to drive [while] interacting with anything, for safety.” In the backseat, passengers like P2.3 reciprocated driver needs to keep the conversation comfortable and topically appropriate, noting how “in a car there is that dynamic of: they don’t want to make you feel uncomfortable, and if I just bring up a topic that makes them feel uncomfortable, I don’t think that [helps]”. The highly socially-conscious passenger P2.2 acknowledged:

“a fine line between having a genuine interest and prying [...] as passengers, we have to be conscious of the fact that these drivers don’t want to offend us, because after all, their rating is at stake, so questions should not be intrusive”

In addition to fears of crossing personal boundaries of drivers, passengers also took care to **minimize chances of belittling or criticizing drivers**, out of awareness for their vulnerabilities: P1.2 “*don’t really feel that it’s like right for me to criticize the way they’re doing their job, because I don’t really know anything about what it’s like being a rideshare driver*”. P2.2 reflected on their own relative power as a consumer more explicitly: “*it is important that we recognize what our position is in the rideshare thing, and we don’t offend people or put them on spots*”.

To help them approach heavier topics, passengers (P1.4, P2.4, P3.2) found **DRIVEN** effective at presenting thought-provoking content (e.g., harassment or family obligations) in creative and memorable ways – “*fun to play in the backseat [...] I was pretty invested*” (D2.2, D3.1) – but it can also be too long, heavy or “*emotionally taxing*” for a single ride (P2.4). For instance, D2.2 “*worry about it being too much for some passengers, [since] not everybody [is] in the mood for deep [content] or or even hearing stories*”. However, passengers felt more comfortable discussing **lighter, ridehail-specific logistics**, including the number of rides the driver completed and reasons for cancellations (P3.3), time spent between rides (P2.2) or how ratings compare across service platforms (P2.3). When combined captivating game mechanics, more casual topics also help alleviate driver responsibilities to lighten the mood, by drawing passenger attention away from pressing factors (traffic, pressures of arriving on-time to destination): “*it passes time fast, also especially helpful on short trips or when we are stuck in traffic, riders stop checking how long it’s taking*” - (D3.2). However, highly engaging or competitive gaming environments might also risk upsetting or stressing out passengers who desire a more relaxed environment – D3.1 and D2.2 consider how “*not everyone wants to think hard during a ride, some folks just wanna zone out or scroll their phone and if someone gets questions wrong over and over, it might actually stress them out*”, highlighting priorities to **avoid overly stressful or challenging game mechanisms** that are “*too competitive, too loud, or makes someone feel dumb*”.

6.3.2 Motivating Socially Responsible Passenger Engagement. Instead of expecting increasingly higher qualities of service, we discuss alternative ways that passengers and drivers desire to engage. Topically, these include conversations that discuss personal interests, local cultures or logistical labor. Other incentives that motivate passenger engagement include monetary contributions to drivers or charitable causes, as well as more punitive rating mechanisms that admonish passengers for irresponsible consumption behaviors.

Connecting via Driver Interests, Labor & Local Culture. Passengers yearned to connect with drivers and local events, pointing to in-ride content that could surface such knowledge. For instance, P1.1 recalled how a photo once sparked a conversation with a driver, and proposed physical cards featuring “*little things that maybe a rider could connect to and be like, ‘Oh, cool, you also watch Love Island.’*” Drivers were generally enthused to share ridehail-specific insights (e.g., “*Uber drivers make playlists just for open silences*”) and personal information (e.g., “*Did you know your driver once drove a pet pig to a party?*”) as well as music playlists (D5.2, D3.1); a few also suggested tailoring “*to rider interest or local culture*” (D2.2)

“*Seattle has the most coffee shops in the US, [so it’s] not only about drivers, but also about local city parts – weird, but true [facts]. Or we can have brainy or historical thing here, ridiculous laws*” (D3.1)

While **TICKING ROADS** simulates the challenges of navigating to pick up passengers, drivers suggested other aspects of their labor as gameplay content. For instance, D1.2 suggested gamifying micro-decisions such as “*declining*”

rides that aren't good", which can improve player competency at identifying exploitative rides. D2.2 envisioned games where the player acts as the driver to guess a passenger's mood, to foreground emotional labor that drivers perform: "guess the rider's mood game based on small clues: the way they greeted you, or a storytelling round where the driver shares one situation and the rider has to react [...] so it helps highlight how much reading the room is part of what we do".

Incentive Mechanisms. In terms of motivating further engagement and play, passengers expressed a desire for more tangible ways to act and help alleviate driver labor and working conditions. For instance, even though Octopus tablets can display driver profile information, P1.1 and P1.4 suggested additional ways of presenting surface-level information about drivers to spark conversation. Beyond knowledge-sharing, passengers like P2.2 proposed that directing their winnings to drivers (or charitable causes) would incentivize their engagement: "I would think there should be an option in there where you could play and win something for your driver." Drivers like D5.3 suggested mechanisms that make "the games more interesting, by adding a leaderboard" which could even be extended across platforms: "a leaderboard for the scores – it would even be more challenging if you could do Uber against Lyft" (D2.1). In addition to positive rewards, P1.1 also considered the use of penalties to dissuade poor and rude passenger behaviors, such as giving greater weight to instances when drivers rate passengers poorly:

*If you're consistently reported as a rude a rider by drivers maybe you have to pay a fine [...so riders think]:
Oh, I should probably not be rude to this driver, because then my next ride is gonna cost me more (P1.1)*

7 DISCUSSION

More than a decade ago, Kittur et al. [85] posed the question of whether we can "foresee a future a crowd workplace in which we would want our children to participate?" Borrowing this lens of healthier (crowd) workplaces for posterity, we discuss implications for future designers of algorithmically-mediated services, renewed approaches to advocacy and directions for how future technology interventions can support workers and their allies.

7.1 Design Implications for Algorithmically-Mediated Production and Creation

Our study surfaced opportunities for new passenger-driver interactions and incentives that can promote more aligned and mutually beneficial behaviors between drivers and passengers (§6.3.2). However, advancing and realizing worker-consumer solidarity in practice – e.g., socially responsible consumption, multi-stakeholder unity and altered platform design decisions – will require core changes to how we all perceive and manage these services, at multiple scales. At the level of individual service interactions, designers might consider tacit ways of reminding consumers to engage in more aware, intentional and responsible micro-decisions that carry managerial function [7] – e.g., *whether to tip drivers before or after a ride* – we suggest in §7.3 future directions for training drivers to resist harmful micro-management by algorithms. More collectively, consumers can begin looking for more local means of advocating and influencing protective driver regulations [59] – described further in §7.2.

Examining Consumer Interactions to Visibilize Labor. Ma et al. [102] showed how platform designers experience internal conflict that arise from value differences between their moral identities and platforms' business objectives, corroborating the claim by Giesler et al. [52] that platforms intentionally cultivate expectation-setting consumer behaviors such as tipping. Here we observe a triadic power relationship between algorithmic managers (or AMs [141]), consumers and workers (ordered from high to low) [60], which can inform the design of an entire ecosystem of algorithmically-mediated services. With the exception of Maffie [104], we are unaware of previous lines of research

within the US that conducted grounded, qualitative analysis on the consumer-AM side of this triangle, which can inform future designs of human-AI co-management or co-creation in general. We encourage future work to analyze more closely the sociomaterial working relations of algorithmically-mediated labor to better document and manage (in)visible accountabilities in labor [144].

Towards Dignified, Autonomy-Preserving & Fulfilling Work. Platforms were heralded for their affordances of flexible and independent work. Much prior research attempted to measure and improve workers' degraded senses of job satisfaction [4] and well-being [113, 169] in the gig economy. But underlying these surface level objectives are missing but central motivations of autonomy, dignity [65] and agency [109] – the lack of which can cause enormous psychological harm. How to design more purposive and negotiable platform mechanisms that satisfy intrinsic worker fulfillment while creating meaning at work? DW5 members mentioned fears of displacement by autonomous vehicles [133]: what are underlying human motivations for resisting them [112]? Can AVs fulfill core service needs of flexibility or convenience? We took care to use the term *ridehail* rather than *rideshare* to (1) emphasize the lack of symmetric power-sharing and (2) bridge the gap between US-based and global scholars [54, 60, 88, 149]. Historically, taxi services evolved from for-hire horse-drawn carriages in the 1600's [41]. But while traditionally licensed taxi operators owned fleets of taxicabs, does this driver-carriage-rider metaphor fit ridehail services?

7.2 Mobilizing Public Discourse and Consumer-led Labor Advocacy through End-User Auditing

“Consumers’ sympathies have not yet been decisively won by platforms or yet lost to the cause of improving gig work conditions” – Healy and Pekarek [59]

Crucially, ridehail offered a unique context of study where the consumer and service-provider are physically and temporally co-located to engage in interpersonal interactions, and our study offer the first attempt within HCI and CSCW to explore possibilities of reshaping the passenger-consumer interaction within a ride [60, 137]. Since consumers as a collective carry significant political power in influencing platform decisions and policy [60], we expand the locus of resistance [24] to “activate” a method “soft action” – gamified interaction [124], with an objective to motivate behavioral change in how consumers respond to, use and resist disruptive technology. By engaging consumers in labor advocacy, we open up opportunities for bottom-up end-user audits [38, 81, 93, 98] (see §7.3 for technical possibilities) or designs (e.g., nudges, asymmetric paternalism) that encourage behaviors of collective resistance more intentional choices (e.g., relatively socially conscious, sustainable competitor alternatives [108, 167]) or non-use [158].

Despite the presence of competitors, platform choices in collude can still inhibit regulatory progress (such as in the case of Prop 22 [149]) – this is the juncture where consumer-led refusal is critical [158]. Compared to the revenue-oriented interactions presented by Play Octopus, our prototypes carry additional purpose of raising awareness around driving conditions. Instead of relying on external ads as a source of support – which harm passenger engagement (e.g., P1.1) – many passengers (e.g., P2.2) inquired about potential prizes and rewards. During workshops, we informally introduced the idea of embedding and promoting local businesses, who can offer direct prizes (e.g., giftcards) and incentives (e.g., coupons) that in turn benefit their operations – enabling more local (and thus scalable) resistance [119].

Legally, tablets like Play Octopus cannot be formally banned from ridehail vehicles [73], since they provide drivers additional earnings when enough passengers engage. Depending on the region, regulations of their presence could put platforms in danger of misclassification lawsuits. However, platforms' intentional emphasis on treating workers as “independent contractor” users (or prosumers [156]) also means that regulatory attention can turn towards consumer protections to advance driver rights [87]. As ridehail internationalizes [54, 60, 88, 149], how should wide-spanning

academic (e.g., computing, labor, organizational management) and geographic communities remain in conversation with one another? As labor increasingly modularize [153], what are more opportunities for more democratic policy creation [91] that allow workers to liberally voice concerns and share insights of working conditions [70] across generations, without fears of platform retaliation?

7.3 Connect Local Communities & Build Meta-Cognitive Abilities with Power-Balanced, Immersive & Creative Technologies

In addition to potentials for engaging collective influence, we discuss how lightweight future technologies can help build more sustainable and connected gig communities, as well as ways to augment more meta-cognitive human skills and intuition so drivers (and workers more broadly) can better detect, monitor and conquer algorithmic management.

Lighter, More Immersive Interactions to Surface Labor & Connect Communities. Our study uncovered driver desires for more in-ride interactions that better integrate passengers with ridehail-specific knowledge while connecting them to local facts and events (§6.2.1). Beyond connecting passengers to immediate physical realities (e.g., traffic, weather), drivers also appreciated how immersive simulations (e.g., TRIVIA RIDE or TICKING ROADS) surfaced (exploitative) labor practices, which persist despite empirical evidence from more than a decade back [123] as well as rising labor consciousness and tech wariness [147]. But while we intended to follow the Tandem Transformational process, our prototypes did not materialize physically, limiting the degree to which we can simulate realistic experiences. Follow up studies might consider how emerging techniques can surface additional driver (and gig) labor at different scales. However, we must preface these ideas by cautioning the reader to first consider potential dangers of applying emerging simulation technologies, what guardrails are necessary? How can mixed or virtual realities [90] simulate individual worker narratives? How can emerging video- and world-generation capabilities capture [16] or re-enact [66] labor accountabilities [62]? What insights can social simulations reveal on collective behaviors [97], or policy formulations [91] of driver communities?

Beyond tablets, passengers expressed preferences for mobile software so they can engage on their own devices. Mobile interactions offer user affordances including social media-like interactions (e.g., discussions [111], chat rooms³) as well as more collection and organization of multimodal (e.g., spatial [56], audio, video) and multi-media data (e.g., new articles, videos, discussions and blogs [24]).

Creatively Engaging Drivers' Meta-Cognitive Capabilities to Promote Driver Dignity. We hope that exposure through immersive gamification can help players (including potential future drivers) to develop behaviors and strategies for resisting manipulative platform tactics [126]. But what are sustainable ways to help drivers regain autonomy, agency and dignity over presently algorithmically-controlled actions? Cameron [22] showed how more resourceful drivers engage in playful acts of resistance, which she termed relational games. Example strategies that drivers employed to win this game included psychological distancing to resist managerial nudges (e.g., notifications, badges) and the accepting of only profitable rides to “make out” with net positive earnings. As tactics of algorithmic management evolve, what are other creative interaction probes and interventions that we can build to train workers' metacognitive abilities (e.g., planning [115], monitoring [138], reflecting [118]) so they can more actively reflect on, track and resist mechanisms of algorithmic control?

³E.g., the weekly X space hosted by Drivers Coast 2 Coast

8 CONCLUSION

This work demonstrated the possibility for playable in-ride interventions to motivate consumer-led advocacy for labor conditions of ridehail drivers. We contribute a novel direction for worker advocacy that bridges the computing and labor communities. Future work can draw upon its insights to (1) gain more in-depth understanding of algorithmically-managed labor relations and organizations (2) foster and mobilize scalable consumer solidarity and advocacy (3) build creative interventions that train human workers to resist harmful algorithmic control.

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9 APPENDIX

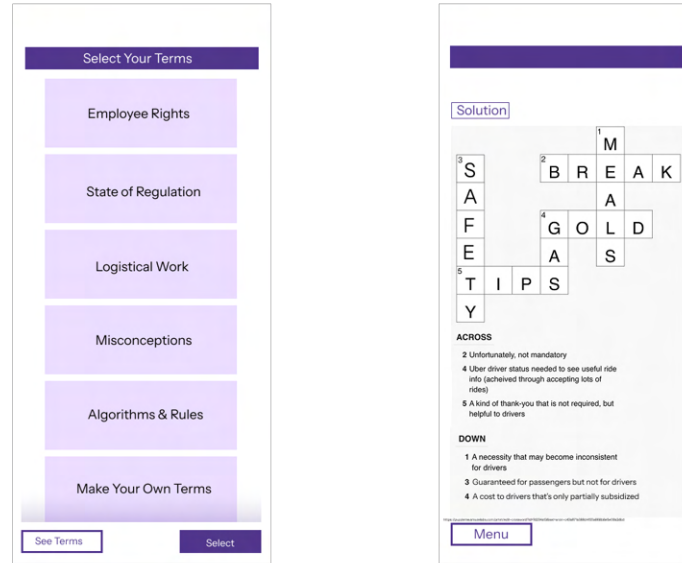


Fig. 8. **CrossRoads** contains driver-selected ridehail concepts

9.1 CrossRoads

Resembling standard crossword puzzles, **CrossRoads** embeds ridehail-related terms and clues to expose such knowledge to passengers. In addition to incorporating ground-truth ridehail concepts, **CrossRoads** contain mechanisms allowing drivers to pick and define their own terms and clues, affording them agency to select ridehail topics most relevant to their own experiences while improve replayability of the game for passengers across rides. The puzzle nature of the crossword orients players to focus on character order as opposed to the ridehail-related terms, but the small screensize of mobile and tablet devices limits content and therefore potential to introduce intermixing. In this prototype, we embedded concrete definitions of concepts (fulfilling requirement for ground truth answers) related to driver rights, regulations, algorithmic management strategies, and logistical burdens.

While driver D2.1 saw potential in **CrossRoads** as a “good distraction” from work for riders, others found it “boring”, cognitively demanding, and criticized its lack of a “social loop” to interact with the driver, as well as in failing to contribute to “energy I’d want in a fun way” - D2.2. Combining these concerns with the difficulties of typing on a tablet keyboard, we decided to eliminate this prototype after the second driver workshop.



Fig. 9. **Dilemmas @ Work** contain black cards representing (ridehail) work dilemmas & white cards with actions to take in response

9.2 Dilemmas @ Work

Based off of the popular social party game *Cards Against Humanity* and inspired by related applications of the card game towards discussion of contexts such as AI ethics [161], as well as driver-led advocacy [116], we prototyped **Dilemmas @ Work**, which adapted the black card deck to represent work dilemmas that drivers and traditional workers might face in their everyday labor – leveraging intermixing (§3.2.1). Correspondingly, white cards depicted potential strategies for handling the various dilemmas presented in black. Designed for a physical social context, the random dealing of cards each round creates a replayable (§3.1.1) experience even among the same group of players. Meanwhile, the objective of humoring the dealing player of each round serves to obfuscate the concepts ridehail vulnerabilities.

Participants of the first driver workshop ranked **Dilemmas @ Work** as the lowest among presented prototypes, explaining how its design to operate outside of a ride discourages engagement with the topic: “*I don’t know that I would see many people actually doing it, if the purpose of this is to educate riders on the driver experience*”, especially since they believed “*the impetus for any of these games would be [with how they are played] during a ride*” - DW1.1.

9.3 Passenger Protocol

- (1) **[Character Card Activity]** Let’s start off by sharing how long you’ve been a rider, how frequently you ride, what platform(s) you use, and where?
- (2) What do you know about rideshare?
 - (a) What aspects of rideshare driving do you know of that are maybe surprising or concerning?
 - (b) Other than driving, what other responsibilities do you think drivers have when working for their platform?
 - (c) Reflecting on your experiences as a passenger, what rides have you taken that are most memorable? Did you learn anything about rideshare driving that was interesting or surprising?
 - (d) What makes you tip a driver more, or give higher, 5-star ratings?
 - (e) On the other hand, what types of rides make you most stressed out or annoyed?

(3) Introducing Game Prototypes

(a) *Ranking Game Heuristics*

- (i) Rank the games based on which you thought were the most fun. How you define fun is up to you, but please explain your rationale.
- (ii) Rank the games based on whether you could see yourself(selves) playing more than once?
- (iii) Please rank the games based on how well-integrated you thought the concepts of ridesharing were (most sneaky)? Would you have noticed the concepts if we didn't point them out? How did it impact the gameplay?
- (iv) Rank based on how easy the games would be to play in an uber ride – potentially on an Octopus-like tablet
- (v) Rank based on the amount of collaboration/interaction you could have/would want to have with the driver in each game.
- (vi) Rank based on which games are the most casual/easiest to pick up/stop at any point – easiest on the mind.
- (vii) Rank each one based on how likely you would be to recommend the games to a friend?
- (viii) Rank which ones make you more curious about rideshare driving conditions. Which ones serve as a launch point and make you want to learn more about them?

(b) *Embedded Concepts*

- (i) Does this game capture the rideshare
- (ii) What other aspects of ridesharing are you curious about? Could you imagine it as a part of the games?
- (iii) Last row are for important concepts that don't yet fit with the existing game prototypes concept(s) accurately?

(4) Reflections

- (a) Other than gamified experiences, what are other interactive modes of communicating driving conditions that you'd be receptive to?
- (b) Were there any particular concepts that would be better implemented in this format rather than in a game?
- (c) Take this time to revisit your character card rankings from the beginning, how did these change?
- (5) Which driving conditions issues are you most curious or concerned about?

9.4 Driver Protocol

(1) Why rideshare?

- (a) Why did each of you start rideshare, and how long have you been driving?
- (b) What are you hoping to learn or gain from the workshop today (can be from us or the other drivers)?
- (c) Now (as a group) choose the question that you would all like to share with passengers the most. Why this question?
- (d) How often do riders talk to you about these with you?
- (e) By contrast, which issues are not necessarily good ideas to be discussed with passengers or in a game?

(2) Introducing Game Prototypes

- (a) Does this game capture the rideshare concept(s) accurately?
- (b) What other aspects of ridesharing can you imagine this game to include?

- (c) How does your experience shape the way you ordered the pieces of information?
- (d) Would you want to subject passengers to similar decisions or experiences (even if simulated) as you?
- (e) Are you interested in knowing more about drivers who identify differently than you? Why [what insights would this tell you if you knew this]?

(3) Reflections

- (a) Jot down your thoughts, concerns and additional ideas about these proposed gaming experiences
- (b) Choose the concern that you think would be the biggest blocker to riders adopting this game. How come?